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DATE:1/7/2026

Set 1: Monthly Sales Data

Dataset: Monthly Sales

1a. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

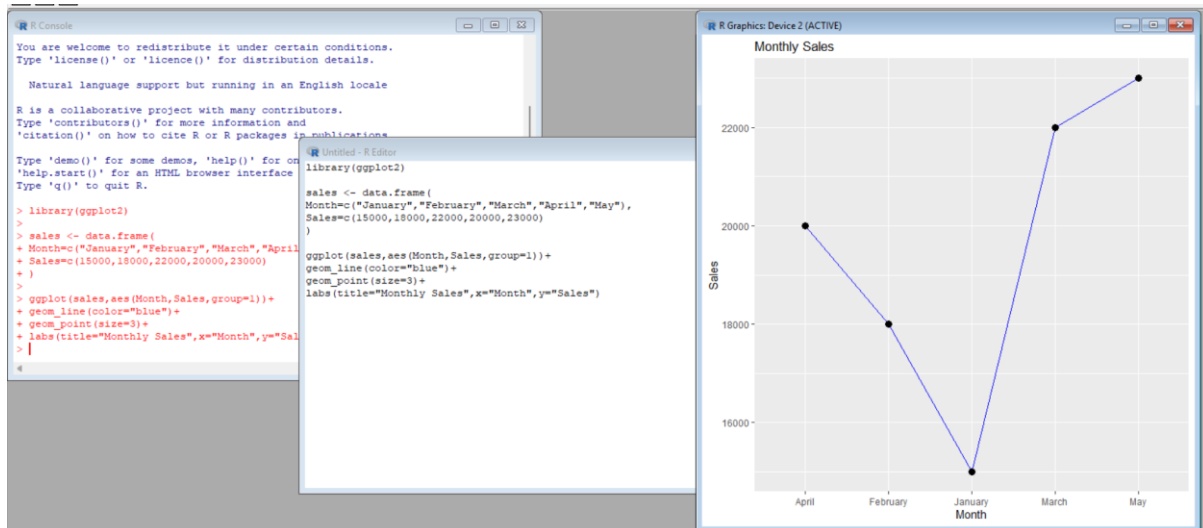
Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000
May	23000

```
1a. library(ggplot2)

sales <- data.frame(
  Month=c("January","February","March","April","May"),
  Sales=c(15000,18000,22000,20000,23000)
)

ggplot(sales,aes(Month,Sales,group=1))+
  geom_line(color="blue")+
  geom_point(size=3)+
  labs(title="Monthly Sales",x="Month",y="Sales")

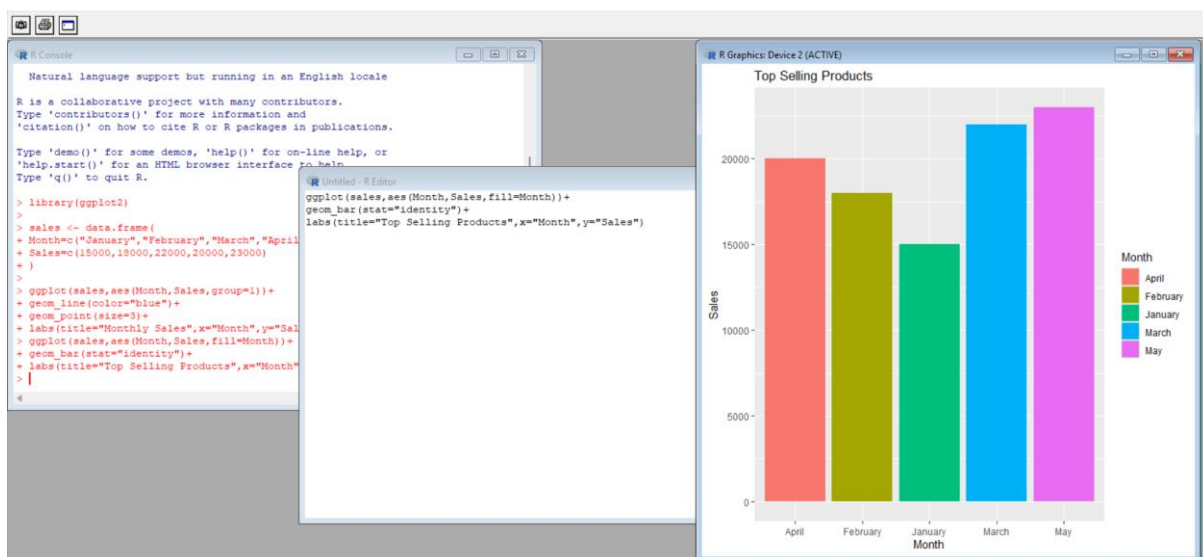
output:
```



1b. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

```
ggplot(sales, aes(Month, Sales, fill=Month)) +
  geom_bar(stat="identity") +
  labs(title="Top Selling Products", x="Month", y="Sales")
```

output:

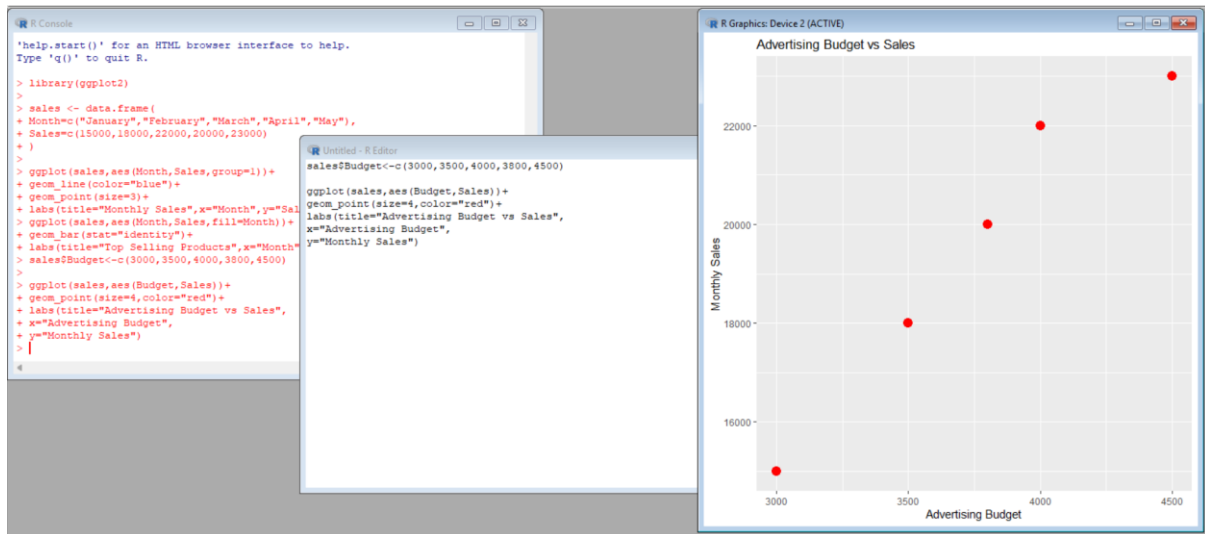


1c. Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

```
sales$Budget<-c(3000,3500,4000,3800,4500)
ggplot(sales, aes(Budget, Sales)) +
  geom_point(size=4, color="red") +
  labs(title="Advertising Budget vs Sales",
  x="Advertising Budget",
```

```
y="Monthly Sales")
```

output:



1d. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

```
library(shiny)

sales <- data.frame(

  Month = c("January", "February", "March", "April", "May"),

  Sales = c(15000, 18000, 22000, 20000, 23000)

)

ui <- fluidPage(

  titlePanel("Monthly Sales Dashboard"),

  sidebarLayout(

    sidebarPanel(

      selectInput("chart",

        "Choose Chart:",

        choices = c("Line Chart",

          "Bar Chart"))

    ),

    mainPanel(

      plotOutput("salesPlot")

    )

  )

)
```

```

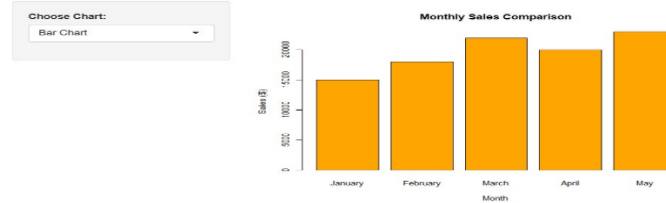
    )
  )
  server <- function(input, output){

    output$salesPlot <- renderPlot({
      if(input$chart == "Line Chart"){
        plot(sales$Sales,
              type = "o",
              xaxt = "n",
              xlab = "Month",
              ylab = "Sales ($)",
              main = "Monthly Sales Trend",
              col = "blue",
              lwd = 2)
        axis(1,
              at = 1:nrow(sales),
              labels = sales$Month)
      }else{
        barplot(sales$Sales,
                 names.arg = sales$Month,
                 col = "orange",
                 xlab = "Month",
                 ylab = "Sales ($)",
                 main = "Monthly Sales Comparison")
      }
    })
  }
}

output:

```

Monthly Sales Dashboard



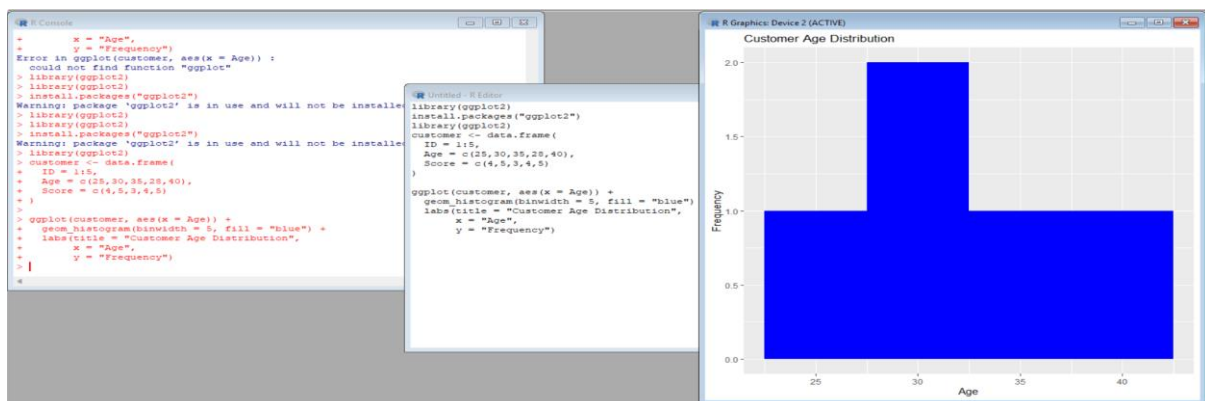
Set 2. Customer Feedback Analysis

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

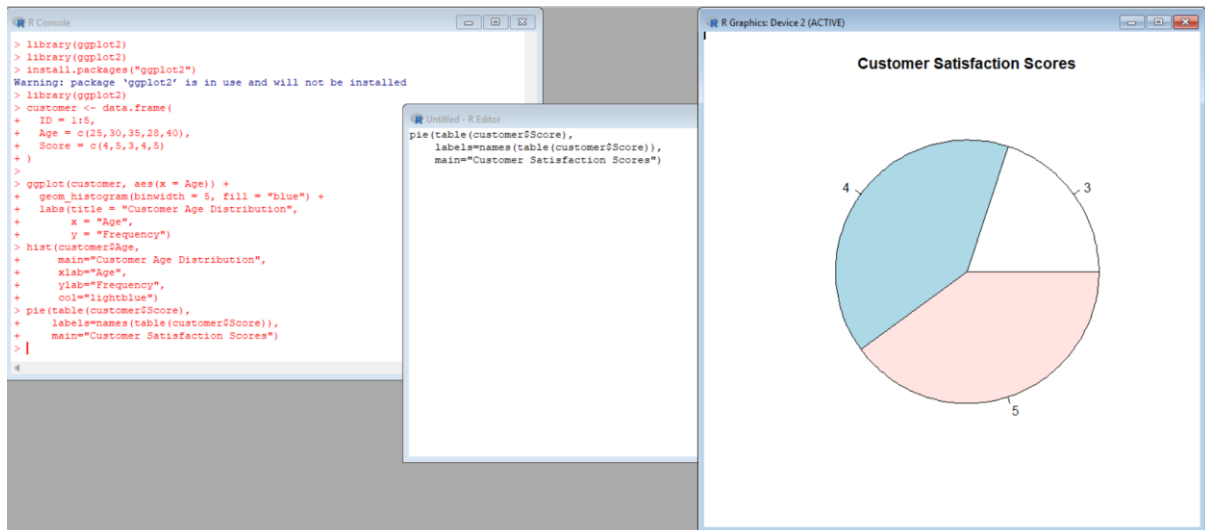
2a. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
A. hist(customer$Age,
      main="Customer Age Distribution",
      xlab="Age",
      ylab="Frequency",
      col="lightblue")
```



2.b. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

```
A. pie(table(customer$Score),
      labels=names(table(customer$Score)),
      main="Customer Satisfaction Scores")
```

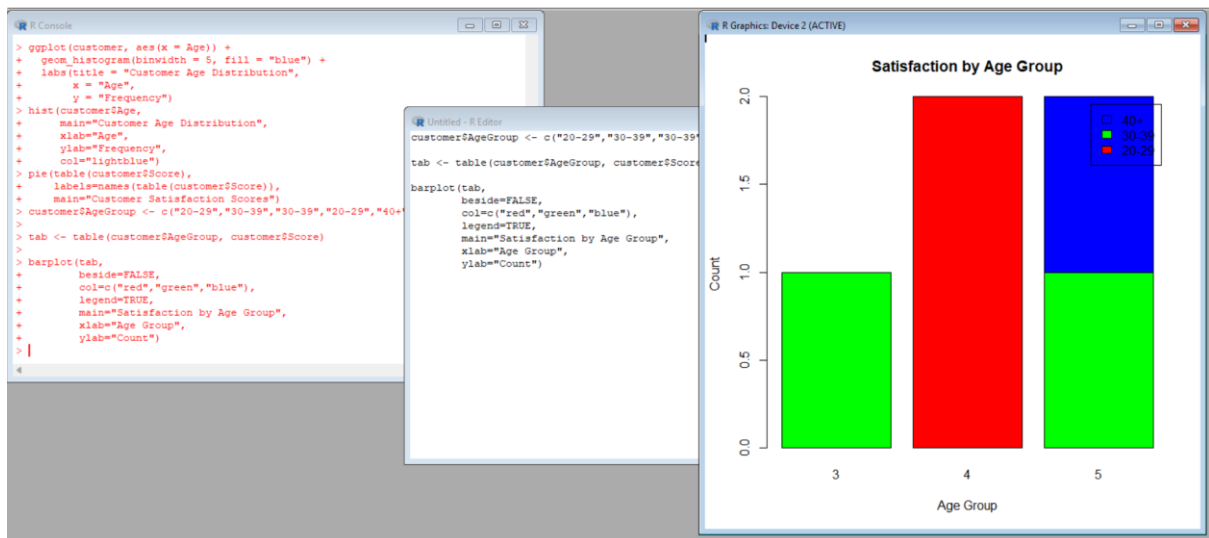


2c. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
customer$AgeGroup <- c("20-29","30-39","30-39","20-29","40+")
```

```
tab <- table(customer$AgeGroup, customer$Score)
```

```
barplot(tab,
      beside=FALSE,
      col=c("red","green","blue"),
      legend=TRUE,
      main="Satisfaction by Age Group",
      xlab="Age Group",
      ylab="Count")
```



2d. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
install.packages("wordcloud")
```

```
library(wordcloud)
```

```
feedback <- c("Good", "Excellent", "Good", "Fast", "Support",
```

```
              "Excellent", "Good", "Friendly", "Quality", "Fast")
```

```
wordcloud(feedback)
```

```
title("Customer Feedback Word Cloud")
```

Customer Feedback Word Cloud

good

Set 3: Employee Performance Evaluation

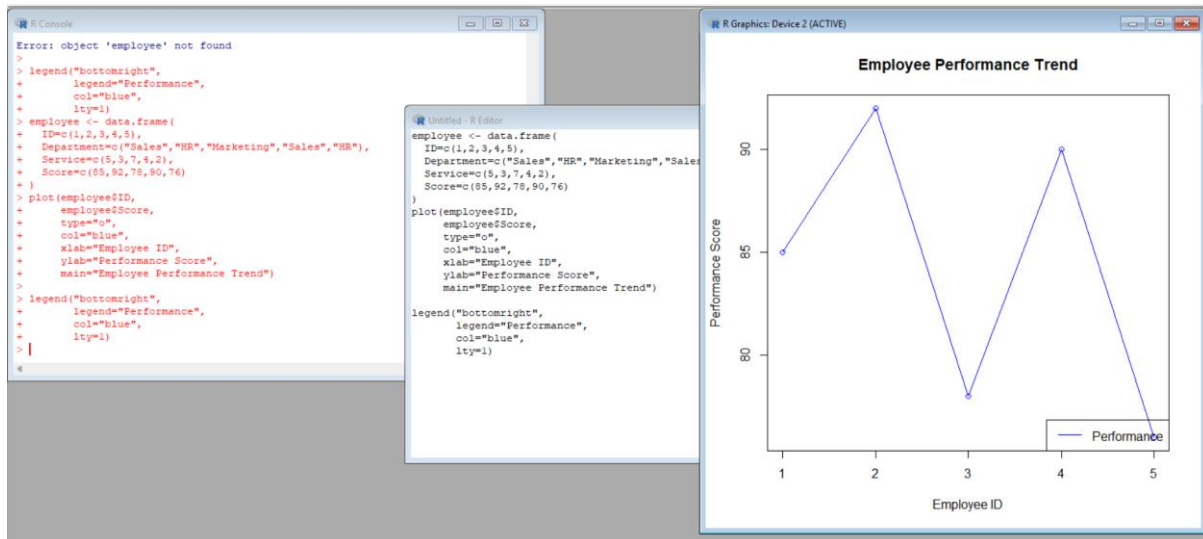
Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

3a. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

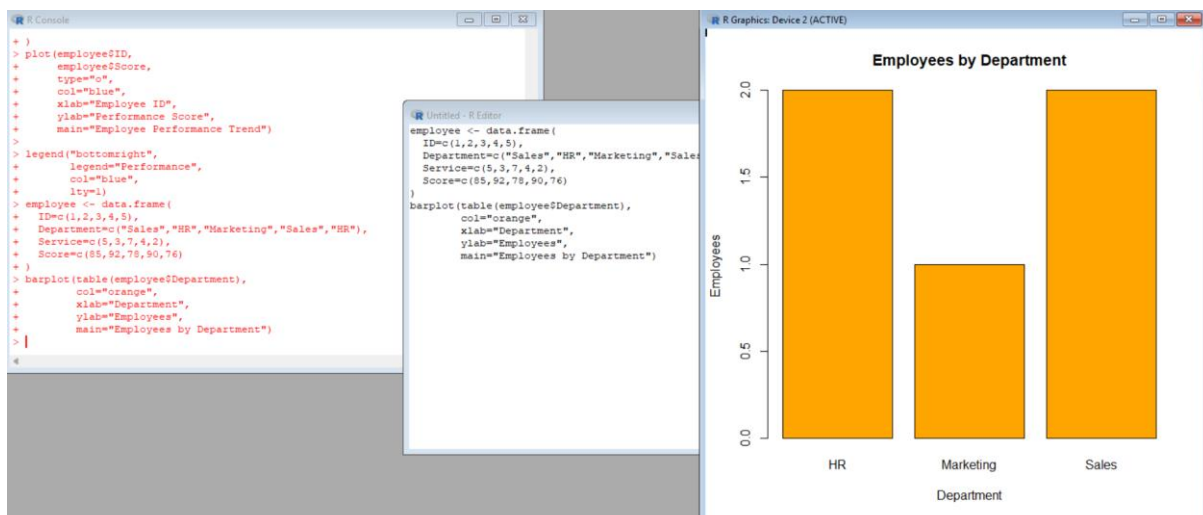
```
employee <- data.frame(
  ID=c(1,2,3,4,5),
  Department=c("Sales","HR","Marketing","Sales","HR"),
  Service=c(5,3,7,4,2),
  Score=c(85,92,78,90,76)
)
```

```
plot(employee$ID,
      employee$Score,
      type="o",
      col="blue",
      xlab="Employee ID",
      ylab="Performance Score",
      main="Employee Performance Trend")
legend("bottomright",
      legend="Performance",
      col="blue",
      lty=1)
```

3b. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

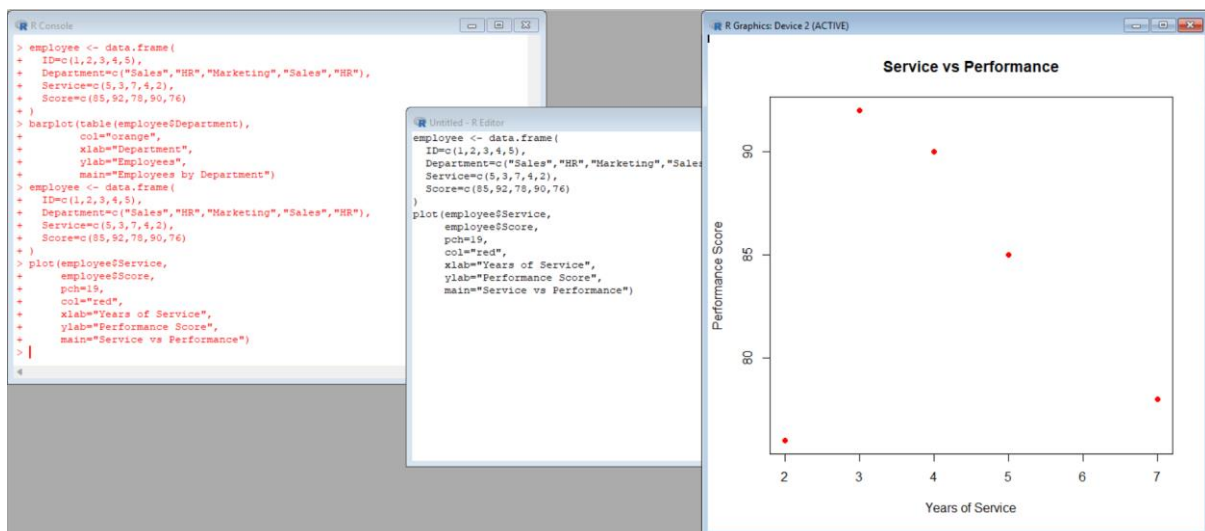
```
employee <- data.frame(
  ID=c(1,2,3,4,5),
  Department=c("Sales","HR","Marketing","Sales","HR"),
  Service=c(5,3,7,4,2),
  Score=c(85,92,78,90,76)
)
barplot(table(employee$Department),
  col="orange",
  xlab="Department",
  ylab="Employees",
  main="Employees by Department")
```



3c. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

```
employee <- data.frame(
  ID=c(1,2,3,4,5),
  Department=c("Sales","HR","Marketing","Sales","HR"),
  Service=c(5,3,7,4,2),
  Score=c(85,92,78,90,76)
)

plot(employee$Service,
  employee$Score,
  pch=19,
  col="red",
  xlab="Years of Service",
  ylab="Performance Score",
  main="Service vs Performance")
```



3d. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

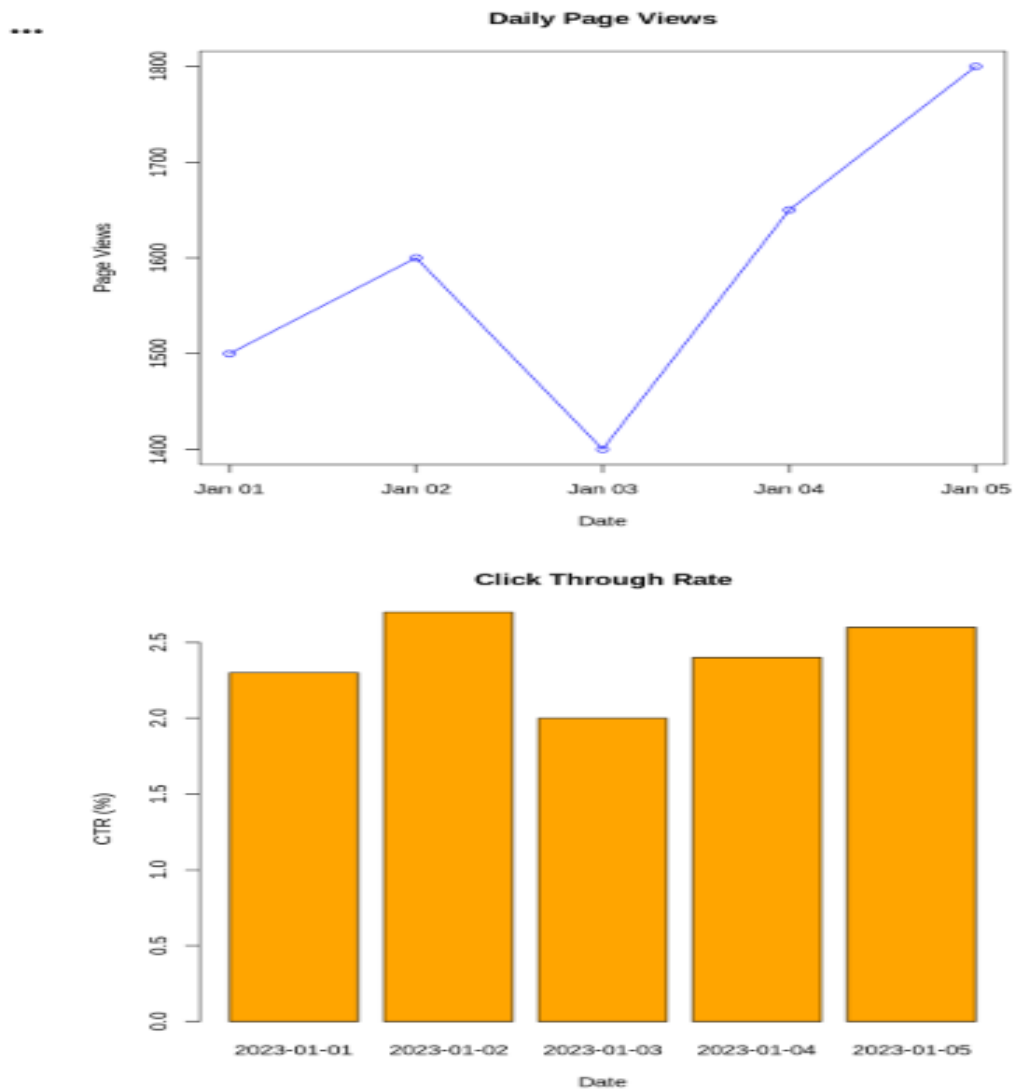
```
traffic <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),
  PageViews = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6)
)

plot(traffic$Date, traffic$PageViews,
```

```

type="o",
col="blue",
xlab="Date",
ylab="Page Views",
main="Daily Page Views")
barplot(traffic$CTR,
names.arg=traffic$Date,
col="orange",
xlab="Date",
ylab="CTR (%)",
main="Click Through Rate")

```



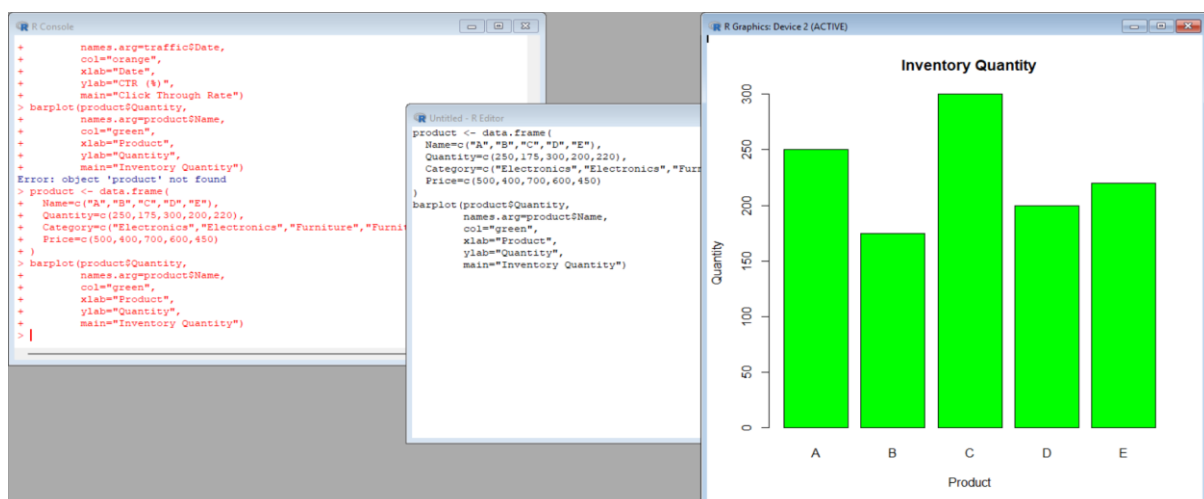
Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

4a. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
product <- data.frame(  
  Name=c("A","B","C","D","E"),  
  Quantity=c(250,175,300,200,220),  
  Category=c("Electronics","Electronics","Furniture","Furniture","Electronics"),  
  Price=c(500,400,700,600,450)  
)  
barplot(product$Quantity,  
  names.arg=product$Name,  
  col="green",  
  xlab="Product",  
  ylab="Quantity",  
  main="Inventory Quantity")
```

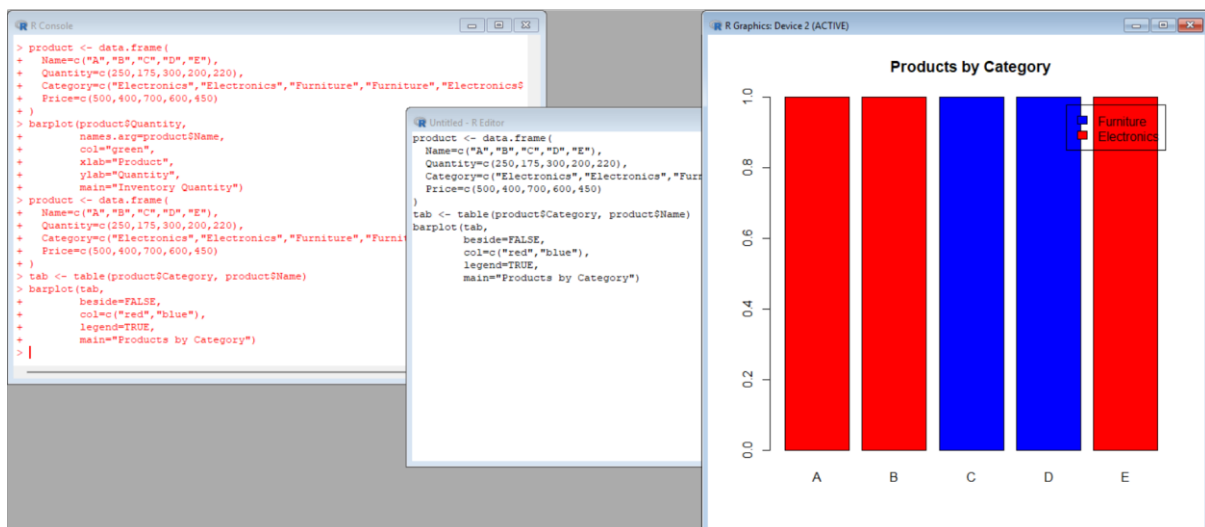


4b. Generate a stacked bar chart to show the quantity of each product within different product categories.

```
product <- data.frame(
  Name=c("A","B","C","D","E"),
  Quantity=c(250,175,300,200,220),
  Category=c("Electronics","Electronics","Furniture","Furniture","Electronics"),
  Price=c(500,400,700,600,450)
)

tab <- table(product$Category, product$Name)

barplot(tab,
  beside=FALSE,
  col=c("red","blue"),
  legend=TRUE,
  main="Products by Category")
```



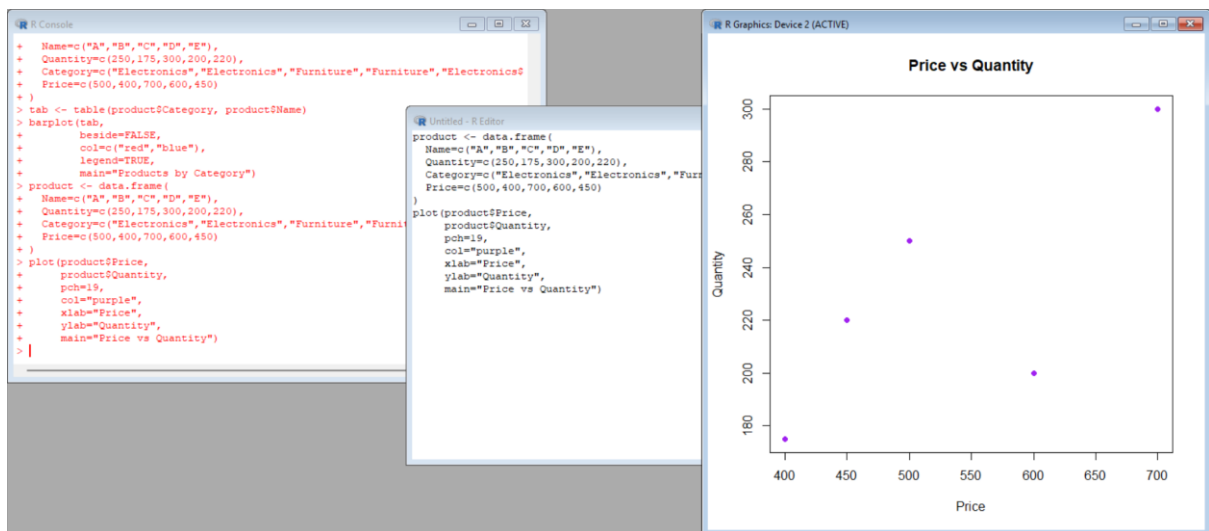
4c. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
product <- data.frame(
  Name=c("A","B","C","D","E"),
  Quantity=c(250,175,300,200,220),
  Category=c("Electronics","Electronics","Furniture","Furniture","Electronics"),
  Price=c(500,400,700,600,450)
)
```

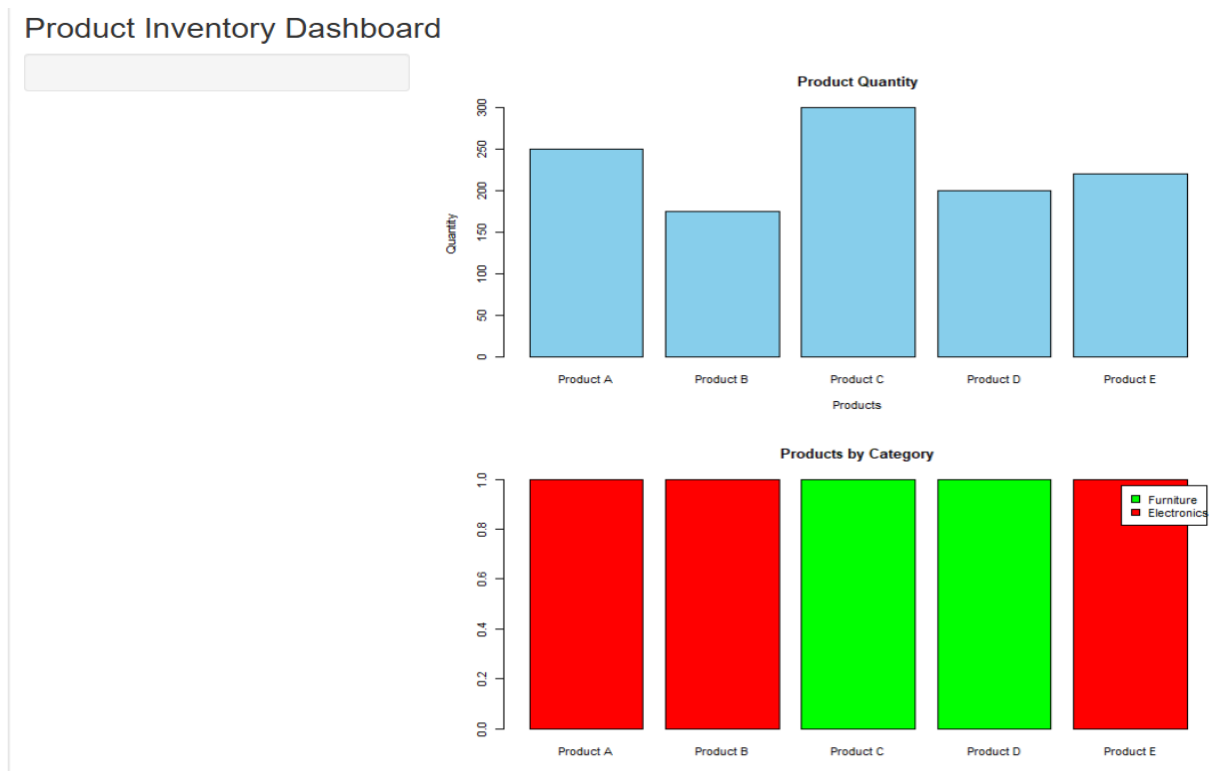
```

plot(product$Price,
      product$Quantity,
      pch=19,
      col="purple",
      xlab="Price",
      ylab="Quantity",
      main="Price vs Quantity")

```



4d. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.



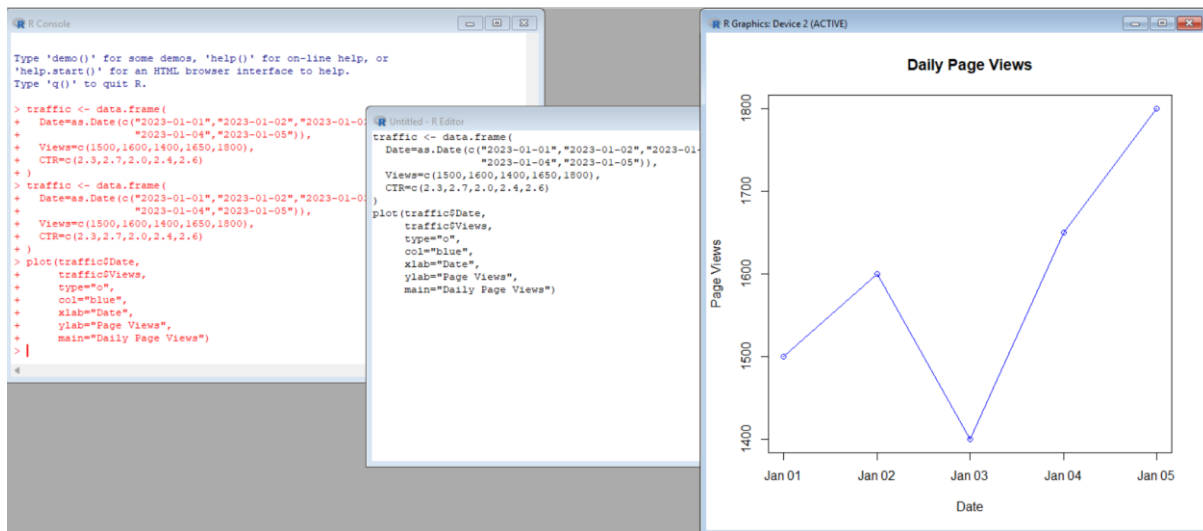
Set 5: Website Analytics

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

5a. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
traffic <- data.frame(  
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03",  
    "2023-01-04","2023-01-05")),  
  Views=c(1500,1600,1400,1650,1800),  
  CTR=c(2.3,2.7,2.0,2.4,2.6)  
)  
  
plot(traffic$Date,  
  traffic$Views,  
  type="o",  
  col="blue",  
  xlab="Date",  
  ylab="Page Views",  
  main="Daily Page Views")
```

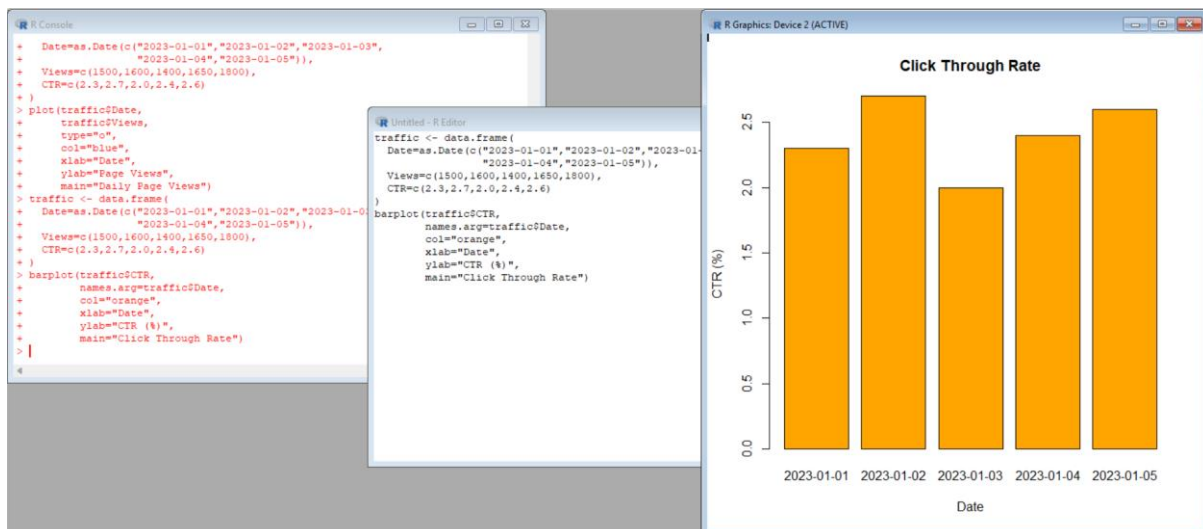


5b. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```

barplot(traffic$CTR,
        names.arg=traffic$Date,
        col="orange",
        xlab="Date",
        ylab="CTR (%)",
        main="Click Through Rate")

```



5c. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

```

traffic <- data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03",
    "2023-01-04","2023-01-05")),

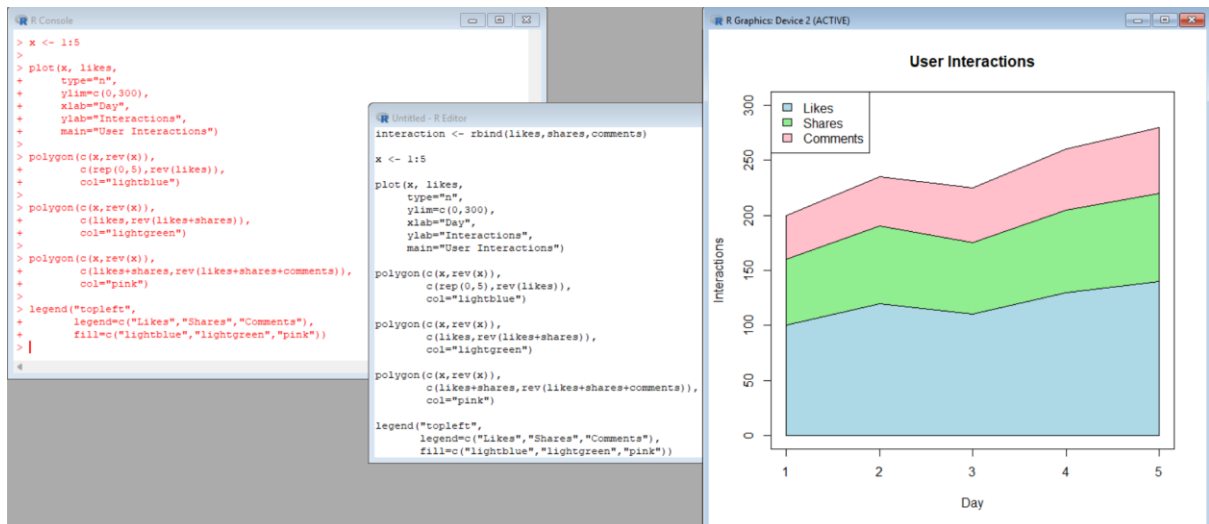
```



```

Views=c(1500,1600,1400,1650,1800),
CTR=c(2.3,2.7,2.0,2.4,2.6)
)
likes <- c(100,120,110,130,140)
shares <- c(60,70,65,75,80)
comments <- c(40,45,50,55,60)
interaction <- rbind(likes,shares,comments)
x <- 1:5
plot(x, likes,
     type="n",
     ylim=c(0,300),
     xlab="Day",
     ylab="Interactions",
     main="User Interactions")
polygon(c(x,rev(x)),
        c(rep(0,5),rev(likes)),
        col="lightblue")
polygon(c(x,rev(x)),
        c(likes,rev(likes+shares)),
        col="lightgreen")
polygon(c(x,rev(x)),
        c(likes+shares,rev(likes+shares+comments)),
        col="pink")
legend("topleft",
      legend=c("Likes","Shares","Comments"),
      fill=c("lightblue","lightgreen","pink"))

```



5d. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Website Traffic Dashboard

